Status	Finished
Started	Tuesday, 27 October 2026, 9:00 PM
Completed	Tuesday, 27 October 2026, 9:04 PM
Duration	4 mins 17 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**
Correct
Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(log n)
- b. O(n log n)
- c. O(1) ✓
- d. O👎

Question **2**
Correct
Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Deleting elements from a list
- b. Determining whether an element is present in a list ✓
- c. Sorting elements in a list
- d. Inserting elements into a list

Question **3**
Correct
Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is unsorted
- b. When the list contains duplicate elements
- c. When the list is sorted ✓
- d. When the list is very small

Question **4**
Correct
Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Both (1) & (3)
- b. Binary ✓
- c. Linear
- d. Hash

Question **5**
Correct
Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Quick search
- b. Linear search ✓
- c. Bubble search
- d. Merge search

Question **6**
Correct
Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the entire list
- b. At the beginning of the list
- c. In the right sublist
- d. In the left sublist ✓

Question **7**
Correct
Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It does not work with negative numbers
- c. It can only be applied to large lists
- d. It is slower than linear search for small lists

Question **8**
Correct
Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 90 and 99 ✓
- b. 89 and 94
- c. 90 and 100
- d. 94 and 99

Question **9**
Correct
Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues from the beginning
- b. The search continues in the left or right sublist ✓
- c. The search stops
- d. The list is sorted

Question **10**
Correct
Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It compares elements sequentially
- b. It works on unsorted lists
- c. It always starts from the beginning of the list
- d. It can be applied only if the list is sorted ✓

Question **11**
Incorrect
Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The search stops
- b. The higher middle element is chosen as the middle element
- c. The middle element is chosen randomly
- d. The lower middle element is chosen as the middle element ✗

Question **12**
Correct
Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is somewhere in the middle of the array ✓
- b. Item is the last element in the array or item is not there at all
- c. When the item is not the array at all
- d. When the item is the last element in the array

Question **13**
Incorrect
Mark 0.00 out of 1.00

In binary search, how is the middle element determined?

- a. By dividing the list length by two
- b. By comparing each element sequentially ✗
- c. By using a hash function
- d. By starting from the first element

Question **14**
Correct
Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 10 ✓
- b. 100
- c. 500
- d. 1000

Question **15**
Correct
Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the left sublist
- b. At the beginning of the list
- c. In the right sublist ✓
- d. In the middle of the list